192425192

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EXP11:

```
#include <stdio.h>
#include <math.h>

int main() {
    int i;
    double log2fact = 0.0;

    /* a) The key is a permutation of 25 letters placed in the 5x5 matrix */
    for (i = 2; i <= 25; i++) log2fact += log2((double)i);
    printf("a) Number of possible keys = 25! = about 2^%.2f  (approx 2^84)\n", log2fact);
    printf("   25! = 15511210043330985984000000\n\n");

    /* b) Matrices that give the same encryption:
          cyclic shift of the 5 rows  (5 ways) and
          cyclic shift of the 5 columns (5 ways)  => 25 equivalent matrices */
    printf("b) Equivalent matrices (row shifts x column shifts) = 5 x 5 = 25\n");
    printf("   Effectively unique keys = 25!/25 = 24! = about 2^%.2f  (approx 2^79)\n",
          log2fact - log2(25.0));
    return 0;
}
```

OUTPUT:

```
a) Number of possible keys = 25! = about 2^83.68  (approx 2^84)
   25! = 15511210043330985984000000

b) Equivalent matrices (row shifts x column shifts) = 5 x 5 = 25
   Effectively unique keys = 25!/25 = 24! = about 2^79.04  (approx 2^79)
```